# Package 'betaSandwich'

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**Title** Robust Confidence Intervals for Standardized Regression Coefficients

**Version** 1.0.9.9000

**Description** Generates robust confidence intervals for standardized regression coefficients using heteroskedasticity-consistent standard errors for models fitted by lm() as described in Dudgeon (2017) <doi:10.1007/s11336-017-9563-z>. The package can also be used to generate confidence intervals for R-squared, adjusted R-squared, and differences of standardized regression coefficients. A description of the package and code examples are presented in Pesigan, Sun, and Cheung (2023) <doi:10.1080/00273171.2023.2201277>.

**URL** https://github.com/jeksterslab/betaSandwich, https://jeksterslab.github.io/betaSandwich/

**BugReports** https://github.com/jeksterslab/betaSandwich/issues

**License** MIT + file LICENSE

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## Contents

BetaADF . . . . . 2
BetaHC . . . . . 4
BetaN . . . . . 6

coef.betasandwich . . . . . 7
coef.essandwich . . . . . 8
confint.betasandwich . . . . . 9
confint.essandwich . . . . . 10
DeltaRSqSandwich . . . . . 11
DiffBetaSandwich . . . . . 12
FSqSandwich . . . . . 13
nas1982 . . . . . 14
PCorSandwich . . . . . 15
print.betasandwich . . . . . 16
print.essandwich . . . . . 17
print.summary.essandwich . . . . . 18
RSqSandwich . . . . . 19
SCorSandwich . . . . . 20
summary.betasandwich . . . . . 21
summary.essandwich . . . . . 22
vcov.betasandwich . . . . . 23
vcov.essandwich . . . . . 23

**Index** **25**

---

`BetaADF` *Estimate Standardized Regression Coefficients and the Corresponding Sampling Covariance Matrix Using the Asymptotic Distribution-Free Approach*

---

## Description

Estimate Standardized Regression Coefficients and the Corresponding Sampling Covariance Matrix Using the Asymptotic Distribution-Free Approach

## Usage

```
BetaADF(object, alpha = c(0.05, 0.01, 0.001), std_y = TRUE, std_x = NULL)
```

## Arguments

| | |
|---|---|
| `object` | Object of class `lm`. |
| `alpha` | Numeric vector. Significance level $\alpha$. |
| `std_y` | Logical. Standardize $y$. With `std_y = FALSE`, $y$ is left in its original units ($b_j\sigma_{x_j}$). |
| `std_x` | Character vector or `NULL`. Names of the regressors to standardize. `NULL` (default) standardizes all regressors; `character(0)` standardizes none ($b_j/\sigma_y$). Leaving a dummy variable out keeps its 0-1 contrast. The delta method accounts for the sampling variability of the standard deviations. |

## Details

Note that while the calculation in `BetaADF()` is different from `betaDelta::BetaDelta()` with `type = "adf"`, the results are numerically equivalent. `BetaADF()` is appropriate when sample sizes are moderate to large (`n > 250`). `BetaHC()` is recommended in most situations.

## Value

Returns an object of class `betasandwich` which is a list with the following elements:

**call** Function call.

**args** Function arguments.

**lm_process** Processed `lm` object.

**gamma_n** Asymptotic covariance matrix of the sample covariance matrix assuming multivariate normality.

**gamma_hc** Asymptotic covariance matrix HC correction.

**gamma** Asymptotic covariance matrix of the sample covariance matrix.

**acov** Asymptotic covariance matrix of the standardized slopes.

**vcov** Sampling covariance matrix of the standardized slopes.

**est** Vector of standardized slopes.

## Author(s)

Ivan Jacob Agaloos Pesigan

## References

Browne, M. W. (1984). Asymptotically distribution-free methods for the analysis of covariance structures. *British Journal of Mathematical and Statistical Psychology*, *37*(1), 62–83. doi:10.1111/j.20448317.1984.tb00789.x

Dudgeon, P. (2017). Some improvements in confidence intervals for standardized regression coefficients. *Psychometrika*, *82*(4), 928–951. doi:10.1007/s113360179563z

Pesigan, I. J. A., Sun, R. W., & Cheung, S. F. (2023). betaDelta and betaSandwich: Confidence intervals for standardized regression coefficients in R. *Multivariate Behavioral Research*. doi:10.1080/00273171.2023.2201277

## See Also

Other Beta Sandwich Functions: `BetaHC()`, `BetaN()`, `DeltaRSqSandwich()`, `DiffBetaSandwich()`, `FSqSandwich()`, `PCorSandwich()`, `RSqSandwich()`, `SCorSandwich()`

## Examples

```
object <- lm(QUALITY ~ NARTIC + PCTGRT + PCTSUPP, data = nas1982)
std <- BetaADF(object)
# Methods ------------------------------------------------------
print(std)
summary(std)
```

```
coef(std)
vcov(std)
confint(std, level = 0.95)
```

---

BetaHC *Estimate Standardized Regression Coefficients and the Corresponding Robust Sampling Covariance Matrix Using the Heteroskedasticity Consistent Approach*

---

## Description

Estimate Standardized Regression Coefficients and the Corresponding Robust Sampling Covariance Matrix Using the Heteroskedasticity Consistent Approach

## Usage

```
BetaHC(
  object,
  type = "hc3",
  alpha = c(0.05, 0.01, 0.001),
  std_y = TRUE,
  std_x = NULL,
  g1 = 1,
  g2 = 1.5,
  k = 0.7
)
```

## Arguments

| | |
|---|---|
| object | Object of class lm. |
| type | Character string. Correction type. Possible values are "hc0", "hc1", "hc2", "hc3", "hc4", "hc4m", and "hc5". |
| alpha | Numeric vector. Significance level $\alpha$. |
| std_y | Logical. Standardize $y$. With std_y = FALSE, $y$ is left in its original units ($b_j\sigma_{x_j}$). |
| std_x | Character vector or NULL. Names of the regressors to standardize. NULL (default) standardizes all regressors; character(0) standardizes none ($b_j/\sigma_y$). Leaving a dummy variable out keeps its 0-1 contrast. The delta method accounts for the sampling variability of the standard deviations. |
| g1 | Numeric. g1 value for type = "hc4m". |
| g2 | Numeric. g2 value for type = "hc4m". |
| k | Numeric. Constant k for type = "hc5" $0 \leq k \leq 1$. |

## Value

Returns an object of class `betasandwich` which is a list with the following elements:

**call** Function call.

**args** Function arguments.

**lm_process** Processed `lm` object.

**gamma_n** Asymptotic covariance matrix of the sample covariance matrix assuming multivariate normality.

**gamma_hc** Asymptotic covariance matrix HC correction.

**gamma** Asymptotic covariance matrix of the sample covariance matrix.

**acov** Asymptotic covariance matrix of the standardized slopes.

**vcov** Sampling covariance matrix of the standardized slopes.

**est** Vector of standardized slopes.

## Author(s)

Ivan Jacob Agaloos Pesigan

## References

Dudgeon, P. (2017). Some improvements in confidence intervals for standardized regression coefficients. *Psychometrika*, *82*(4), 928–951. doi:10.1007/s113360179563z

Pesigan, I. J. A., Sun, R. W., & Cheung, S. F. (2023). betaDelta and betaSandwich: Confidence intervals for standardized regression coefficients in R. *Multivariate Behavioral Research*. doi:10.1080/00273171.2023.2201277

## See Also

Other Beta Sandwich Functions: `BetaADF()`, `BetaN()`, `DeltaRSqSandwich()`, `DiffBetaSandwich()`, `FSqSandwich()`, `PCorSandwich()`, `RSqSandwich()`, `SCorSandwich()`

## Examples

```
object <- lm(QUALITY ~ NARTIC + PCTGRT + PCTSUPP, data = nas1982)
std <- BetaHC(object)
# Methods -----------------------------------------------------------
print(std)
summary(std)
coef(std)
vcov(std)
confint(std, level = 0.95)
```

BetaN *Estimate Standardized Regression Coefficients and the Corresponding Sampling Covariance Matrix Assuming Multivariate Normality*

## Description

Estimate Standardized Regression Coefficients and the Corresponding Sampling Covariance Matrix Assuming Multivariate Normality

## Usage

```
BetaN(object, alpha = c(0.05, 0.01, 0.001), std_y = TRUE, std_x = NULL)
```

## Arguments

| | |
|---|---|
| object | Object of class lm. |
| alpha | Numeric vector. Significance level $\alpha$. |
| std_y | Logical. Standardize $y$. With std_y = FALSE, $y$ is left in its original units ($b_j\sigma_{x_j}$). |
| std_x | Character vector or NULL. Names of the regressors to standardize. NULL (default) standardizes all regressors; character(0) standardizes none ($b_j/\sigma_y$). Leaving a dummy variable out keeps its 0-1 contrast. The delta method accounts for the sampling variability of the standard deviations. |

## Details

Note that while the calculation in BetaN() is different from betaDelta::BetaDelta() with type = "mvn", the results are numerically equivalent. BetaN() assumes multivariate normality. BetaHC() is recommended in most situations.

## Value

Returns an object of class betasandwich which is a list with the following elements:

**call** Function call.

**args** Function arguments.

**lm_process** Processed lm object.

**gamma_n** Asymptotic covariance matrix of the sample covariance matrix assuming multivariate normality.

**gamma_hc** Asymptotic covariance matrix HC correction.

**gamma** Asymptotic covariance matrix of the sample covariance matrix.

**acov** Asymptotic covariance matrix of the standardized slopes.

**vcov** Sampling covariance matrix of the standardized slopes.

**est** Vector of standardized slopes.

## Author(s)

Ivan Jacob Agaloos Pesigan

## References

Dudgeon, P. (2017). Some improvements in confidence intervals for standardized regression coefficients. *Psychometrika*, *82*(4), 928–951. doi:10.1007/s113360179563z

Pesigan, I. J. A., Sun, R. W., & Cheung, S. F. (2023). betaDelta and betaSandwich: Confidence intervals for standardized regression coefficients in R. *Multivariate Behavioral Research*. doi:10.1080/00273171.2023.2201277

## See Also

Other Beta Sandwich Functions: `BetaADF()`, `BetaHC()`, `DeltaRSqSandwich()`, `DiffBetaSandwich()`, `FSqSandwich()`, `PCorSandwich()`, `RSqSandwich()`, `SCorSandwich()`

## Examples

```
object <- lm(QUALITY ~ NARTIC + PCTGRT + PCTSUPP, data = nas1982)
std <- BetaN(object)
# Methods -----------------------------------------------------
print(std)
summary(std)
coef(std)
vcov(std)
confint(std, level = 0.95)
```

---

`coef.betasandwich` *Standardized Regression Slopes*

---

## Description

Standardized Regression Slopes

## Usage

```
## S3 method for class 'betasandwich'
coef(object, ...)
```

## Arguments

| | |
|---|---|
| `object` | Object of class `betasandwich`. |
| `...` | additional arguments. |

## Value

Returns a vector of standardized regression slopes.

## Author(s)

Ivan Jacob Agaloos Pesigan

## Examples

```
object <- lm(QUALITY ~ NARTIC + PCTGRT + PCTSUPP, data = nas1982)
std <- BetaHC(object)
coef(std)
```

---

# coef.essandwich *Estimates for an Object of Class* essandwich

---

## Description

Estimates for an Object of Class essandwich

## Usage

```
## S3 method for class 'essandwich'
coef(object, ...)
```

## Arguments

| | |
|---|---|
| object | Object of class essandwich. |
| ... | additional arguments. |

## Value

Returns a vector of estimates.

## Author(s)

Ivan Jacob Agaloos Pesigan

## Examples

```
object <- lm(QUALITY ~ NARTIC + PCTGRT + PCTSUPP, data = nas1982)
std <- BetaHC(object)
out <- RSqSandwich(std)
coef(out)
```

confint.betasandwich *Confidence Intervals for Standardized Regression Slopes*

## Description

Confidence Intervals for Standardized Regression Slopes

## Usage

```
## S3 method for class 'betasandwich'
confint(object, parm = NULL, level = 0.95, ...)
```

## Arguments

| | |
|---|---|
| object | Object of class betasandwich. |
| parm | a specification of which parameters are to be given confidence intervals, either a vector of numbers or a vector of names. If missing, all parameters are considered. |
| level | the confidence level required. |
| ... | additional arguments. |

## Value

Returns a matrix of confidence intervals.

## Author(s)

Ivan Jacob Agaloos Pesigan

## Examples

```
object <- lm(QUALITY ~ NARTIC + PCTGRT + PCTSUPP, data = nas1982)
std <- BetaHC(object)
confint(std, level = 0.95)
```

confint.essandwich *Confidence Intervals for an Object of Class* essandwich

## Description

Confidence Intervals for an Object of Class essandwich

## Usage

```
## S3 method for class 'essandwich'
confint(object, parm = NULL, level = 0.95, ...)
```

## Arguments

| | |
|---|---|
| object | Object of class essandwich. |
| parm | a specification of which parameters are to be given confidence intervals, either a vector of numbers or a vector of names. If missing, all parameters are considered. |
| level | the confidence level required. |
| ... | additional arguments. |

## Value

Returns a matrix of confidence intervals.

## Author(s)

Ivan Jacob Agaloos Pesigan

## Examples

```
object <- lm(QUALITY ~ NARTIC + PCTGRT + PCTSUPP, data = nas1982)
std <- BetaHC(object)
out <- RSqSandwich(std)
confint(out, level = 0.95)
```

`DeltaRSqSandwich` *Estimate Improvement in R-Squared and the Corresponding Sampling Covariance Matrix*

## Description

Estimate Improvement in R-Squared and the Corresponding Sampling Covariance Matrix

## Usage

```
DeltaRSqSandwich(object, alpha = c(0.05, 0.01, 0.001))
```

## Arguments

| | |
|---|---|
| `object` | Object of class `betasandwich`, that is, the output of the `BetaHC()`, `BetaN()`, or `BetaADF()` functions. |
| `alpha` | Numeric vector. Significance level $\alpha$. |

## Details

The sampling covariance matrix is obtained by the delta method from the sampling covariance matrix of the standardized parameters stored in `object` (normal-theory, asymptotically distribution-free, or heteroskedasticity-consistent). Two or more regressors are required.

## Value

Returns an object of class `essandwich` which is a list with the following elements:

**call** Function call.

**fit** The argument `object`.

**args** Function arguments.

**vcov** Sampling covariance matrix of improvement in R-squared (squared semipartial correlation coefficients).

**est** Vector of improvement in R-squared (squared semipartial correlation coefficients).

**fun** Function used ("DeltaRSqSandwich").

## Author(s)

Ivan Jacob Agaloos Pesigan

## See Also

Other Beta Sandwich Functions: `BetaADF()`, `BetaHC()`, `BetaN()`, `DiffBetaSandwich()`, `FSqSandwich()`, `PCorSandwich()`, `RSqSandwich()`, `SCorSandwich()`

## Examples

```
object <- lm(QUALITY ~ NARTIC + PCTGRT + PCTSUPP, data = nas1982)
std <- BetaHC(object)
out <- DeltaRSqSandwich(std)
# Methods ------------------------------------------------------
print(out)
summary(out)
coef(out)
vcov(out)
confint(out, level = 0.95)
```

---

`DiffBetaSandwich` *Estimate Differences of Standardized Slopes and the Corresponding Sampling Covariance Matrix*

---

## Description

Estimate Differences of Standardized Slopes and the Corresponding Sampling Covariance Matrix

## Usage

```
DiffBetaSandwich(object, alpha = c(0.05, 0.01, 0.001))
```

## Arguments

| | |
|---|---|
| `object` | Object of class `betasandwich`, that is, the output of the `BetaHC()`, `BetaN()`, or `BetaADF()` functions. |
| `alpha` | Numeric vector. Significance level $\alpha$. |

## Value

Returns an object of class `essandwich` (and `diffbetasandwich`) which is a list with the following elements:

**call** Function call.

**fit** The argument `object`.

**args** Function arguments.

**vcov** Sampling covariance matrix of differences of standardized slopes.

**est** Vector of differences of standardized slopes.

**fun** Function used ("DiffBetaSandwich").

## Author(s)

Ivan Jacob Agaloos Pesigan

## See Also

Other Beta Sandwich Functions: BetaADF(), BetaHC(), BetaN(), DeltaRSqSandwich(), FSqSandwich(), PCorSandwich(), RSqSandwich(), SCorSandwich()

## Examples

```
object <- lm(QUALITY ~ NARTIC + PCTGRT + PCTSUPP, data = nas1982)
std <- BetaHC(object)
diff <- DiffBetaSandwich(std)
# Methods ------------------------------------------------------
print(diff)
summary(diff)
coef(diff)
vcov(diff)
confint(diff, level = 0.95)
```

---

FSqSandwich *Estimate Cohen's f-Squared and Local f-Squared and the Corresponding Sampling Covariance Matrix*

---

## Description

Estimate Cohen's f-Squared and Local f-Squared and the Corresponding Sampling Covariance Matrix

## Usage

```
FSqSandwich(object, alpha = c(0.05, 0.01, 0.001))
```

## Arguments

| | |
|---|---|
| object | Object of class betasandwich, that is, the output of the BetaHC(), BetaN(), or BetaADF() functions. |
| alpha | Numeric vector. Significance level $\alpha$. |

## Details

The sampling covariance matrix is obtained by the delta method from the sampling covariance matrix of the standardized parameters stored in object (normal-theory, asymptotically distribution-free, or heteroskedasticity-consistent).

## Value

Returns an object of class `essandwich` which is a list with the following elements:

**call** Function call.

**fit** The argument `object`.

**args** Function arguments.

**vcov** Sampling covariance matrix of Cohen's f-squared (fsq) and the local f-squared of each regressor.

**est** Vector of Cohen's f-squared (fsq) and the local f-squared of each regressor.

**fun** Function used ("FSqSandwich").

## Author(s)

Ivan Jacob Agaloos Pesigan

## See Also

Other Beta Sandwich Functions: `BetaADF()`, `BetaHC()`, `BetaN()`, `DeltaRSqSandwich()`, `DiffBetaSandwich()`, `PCorSandwich()`, `RSqSandwich()`, `SCorSandwich()`

## Examples

```
object <- lm(QUALITY ~ NARTIC + PCTGRT + PCTSUPP, data = nas1982)
std <- BetaHC(object)
out <- FSqSandwich(std)
# Methods ------------------------------------------------------
print(out)
summary(out)
coef(out)
vcov(out)
confint(out, level = 0.95)
```

---

# nas1982 *1982 National Academy of Sciences Doctoral Programs Data*

---

## Description

1982 National Academy of Sciences Doctoral Programs Data

## Usage

```
nas1982
```

## Format

Ratings of 46 doctoral programs in psychology in the USA with the following variables:

**QUALITY** Program quality ratings.

**NFACUL** Number of faculty members in the program.

**NGRADS** Number of program graduates.

**PCTSUPP** Percentage of program graduates who received support.

**PCTGRT** Percent of faculty members holding research grants.

**NARTIC** Number of published articles attributed to program faculty member.

**PCTPUB** Percent of faculty with one or more published article.

## References

National Research Council. (1982). *An assessment of research-doctorate programs in the United States: Social and behavioral sciences*. doi:10.17226/9781. Reproduced with permission from the National Academy of Sciences, Courtesy of the National Academies Press, Washington, D.C.

---

`PCorSandwich` *Estimate Squared Partial Correlation Coefficients and the Corresponding Sampling Covariance Matrix*

---

## Description

Estimate Squared Partial Correlation Coefficients and the Corresponding Sampling Covariance Matrix

## Usage

```
PCorSandwich(object, alpha = c(0.05, 0.01, 0.001))
```

## Arguments

| | |
|---|---|
| `object` | Object of class `betasandwich`, that is, the output of the `BetaHC()`, `BetaN()`, or `BetaADF()` functions. |
| `alpha` | Numeric vector. Significance level $\alpha$. |

## Details

The sampling covariance matrix is obtained by the delta method from the sampling covariance matrix of the standardized parameters stored in `object` (normal-theory, asymptotically distribution-free, or heteroskedasticity-consistent). Two or more regressors are required.

## Value

Returns an object of class essandwich which is a list with the following elements:

**call** Function call.

**fit** The argument object.

**args** Function arguments.

**vcov** Sampling covariance matrix of squared partial correlation coefficients.

**est** Vector of squared partial correlation coefficients.

**fun** Function used ("PCorSandwich").

## Author(s)

Ivan Jacob Agaloos Pesigan

## See Also

Other Beta Sandwich Functions: BetaADF(), BetaHC(), BetaN(), DeltaRSqSandwich(), DiffBetaSandwich(), FSqSandwich(), RSqSandwich(), SCorSandwich()

## Examples

```
object <- lm(QUALITY ~ NARTIC + PCTGRT + PCTSUPP, data = nas1982)
std <- BetaHC(object)
out <- PCorSandwich(std)
# Methods -----------------------------------------------------
print(out)
summary(out)
coef(out)
vcov(out)
confint(out, level = 0.95)
```

---

# print.betasandwich *Print Method for an Object of Class* betasandwich

---

## Description

Print Method for an Object of Class betasandwich

## Usage

```
## S3 method for class 'betasandwich'
print(x, alpha = NULL, digits = 4, ...)
```

### Arguments

| | |
|---|---|
| x | Object of class betasandwich. |
| alpha | Numeric vector. Significance level $\alpha$. If alpha = NULL, use the argument alpha used in x. |
| digits | Digits to print. |
| ... | additional arguments. |

### Value

Prints a matrix of standardized regression slopes, standard errors, test statistics, degrees of freedom, p-values, and confidence intervals.

### Author(s)

Ivan Jacob Agaloos Pesigan

### Examples

```
object <- lm(QUALITY ~ NARTIC + PCTGRT + PCTSUPP, data = nas1982)
std <- BetaHC(object)
print(std)
```

---

## print.essandwich *Print Method for an Object of Class* essandwich

---

### Description

Print Method for an Object of Class essandwich

### Usage

```
## S3 method for class 'essandwich'
print(x, alpha = NULL, digits = 4, ...)
```

### Arguments

| | |
|---|---|
| x | Object of class essandwich. |
| alpha | Numeric vector. Significance level $\alpha$. If alpha = NULL, use the argument alpha used in x. |
| digits | Digits to print. |
| ... | additional arguments. |

### Value

Prints a matrix of estimates, standard errors, test statistics, p-values, and confidence intervals.

### Author(s)

Ivan Jacob Agaloos Pesigan

### Examples

```
object <- lm(QUALITY ~ NARTIC + PCTGRT + PCTSUPP, data = nas1982)
std <- BetaHC(object)
out <- RSqSandwich(std)
print(out)
```

---

## print.summary.essandwich

*Print Method for an Object of Class* `summary.essandwich`

---

### Description

Print Method for an Object of Class `summary.essandwich`

### Usage

```
## S3 method for class 'summary.essandwich'
print(x, ...)
```

### Arguments

| | |
|---|---|
| `x` | Object of class `summary.essandwich`. |
| `...` | additional arguments. |

### Value

Prints a matrix of estimates, standard errors, test statistics, p-values, and confidence intervals.

### Author(s)

Ivan Jacob Agaloos Pesigan

### Examples

```
object <- lm(QUALITY ~ NARTIC + PCTGRT + PCTSUPP, data = nas1982)
std <- BetaHC(object)
out <- RSqSandwich(std)
summary(out)
```

RSqSandwich *Estimate R-Squared and Adjusted R-Squared and the Corresponding Sampling Covariance Matrix*

## Description

Estimate R-Squared and Adjusted R-Squared and the Corresponding Sampling Covariance Matrix

## Usage

```
RSqSandwich(object, alpha = c(0.05, 0.01, 0.001))
```

## Arguments

| | |
|---|---|
| object | Object of class betasandwich, that is, the output of the BetaHC(), BetaN(), or BetaADF() functions. |
| alpha | Numeric vector. Significance level $\alpha$. |

## Details

The sampling covariance matrix is obtained by the delta method from the sampling covariance matrix of the standardized parameters stored in object (normal-theory, asymptotically distribution-free, or heteroskedasticity-consistent).

## Value

Returns an object of class essandwich which is a list with the following elements:

**call** Function call.

**fit** The argument object.

**args** Function arguments.

**vcov** Sampling covariance matrix of R-squared and adjusted R-squared.

**est** Vector of R-squared and adjusted R-squared.

**fun** Function used ("RSqSandwich").

## Author(s)

Ivan Jacob Agaloos Pesigan

## See Also

Other Beta Sandwich Functions: BetaADF(), BetaHC(), BetaN(), DeltaRSqSandwich(), DiffBetaSandwich(), FSqSandwich(), PCorSandwich(), SCorSandwich()

## Examples

```
object <- lm(QUALITY ~ NARTIC + PCTGRT + PCTSUPP, data = nas1982)
std <- BetaHC(object)
out <- RSqSandwich(std)
# Methods ------------------------------------------------------
print(out)
summary(out)
coef(out)
vcov(out)
confint(out, level = 0.95)
```

---

`SCorSandwich` *Estimate Semipartial Correlation Coefficients and the Corresponding Sampling Covariance Matrix*

---

## Description

Estimate Semipartial Correlation Coefficients and the Corresponding Sampling Covariance Matrix

## Usage

```
SCorSandwich(object, alpha = c(0.05, 0.01, 0.001))
```

## Arguments

| | |
|---|---|
| `object` | Object of class `betasandwich`, that is, the output of the `BetaHC()`, `BetaN()`, or `BetaADF()` functions. |
| `alpha` | Numeric vector. Significance level $\alpha$. |

## Details

The sampling covariance matrix is obtained by the delta method from the sampling covariance matrix of the standardized parameters stored in `object` (normal-theory, asymptotically distribution-free, or heteroskedasticity-consistent). Two or more regressors are required.

## Value

Returns an object of class `essandwich` which is a list with the following elements:

**call** Function call.

**fit** The argument `object`.

**args** Function arguments.

**vcov** Sampling covariance matrix of semipartial correlation coefficients.

**est** Vector of semipartial correlation coefficients.

**fun** Function used ("SCorSandwich").

### Author(s)

Ivan Jacob Agaloos Pesigan

### See Also

Other Beta Sandwich Functions: `BetaADF()`, `BetaHC()`, `BetaN()`, `DeltaRSqSandwich()`, `DiffBetaSandwich()`, `FSqSandwich()`, `PCorSandwich()`, `RSqSandwich()`

### Examples

```
object <- lm(QUALITY ~ NARTIC + PCTGRT + PCTSUPP, data = nas1982)
std <- BetaHC(object)
out <- SCorSandwich(std)
# Methods -----------------------------------------------------
print(out)
summary(out)
coef(out)
vcov(out)
confint(out, level = 0.95)
```

---

## summary.betasandwich *Summary Method for an Object of Class* betasandwich

---

### Description

Summary Method for an Object of Class betasandwich

### Usage

```
## S3 method for class 'betasandwich'
summary(object, alpha = NULL, digits = 4, ...)
```

### Arguments

| | |
|---|---|
| object | Object of class betasandwich. |
| alpha | Numeric vector. Significance level $\alpha$. If alpha = NULL, use the argument alpha used in object. |
| digits | Digits to print. |
| ... | additional arguments. |

### Value

Returns a matrix of standardized regression slopes, standard errors, test statistics, degrees of freedom, p-values, and confidence intervals.

### Author(s)

Ivan Jacob Agaloos Pesigan

### Examples

```
object <- lm(QUALITY ~ NARTIC + PCTGRT + PCTSUPP, data = nas1982)
std <- BetaHC(object)
summary(std)
```

---

## summary.essandwich *Summary Method for an Object of Class* essandwich

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### Description

Summary Method for an Object of Class essandwich

### Usage

```
## S3 method for class 'essandwich'
summary(object, alpha = NULL, digits = 4, ...)
```

### Arguments

| | |
|---|---|
| object | Object of class essandwich. |
| alpha | Numeric vector. Significance level $\alpha$. If alpha = NULL, use the argument alpha used in object. |
| digits | Digits to print. |
| ... | additional arguments. |

### Value

Returns a matrix of estimates, standard errors, test statistics, p-values, and confidence intervals.

### Author(s)

Ivan Jacob Agaloos Pesigan

### Examples

```
object <- lm(QUALITY ~ NARTIC + PCTGRT + PCTSUPP, data = nas1982)
std <- BetaHC(object)
out <- RSqSandwich(std)
summary(out)
```

## vcov.betasandwich *Sampling Covariance Matrix of the Standardized Regression Slopes*

### Description

Sampling Covariance Matrix of the Standardized Regression Slopes

### Usage

```
## S3 method for class 'betasandwich'
vcov(object, ...)
```

### Arguments

| | |
|---|---|
| object | Object of class betasandwich. |
| ... | additional arguments. |

### Value

Returns a matrix of the variance-covariance matrix of standardized slopes.

### Author(s)

Ivan Jacob Agaloos Pesigan

### Examples

```
object <- lm(QUALITY ~ NARTIC + PCTGRT + PCTSUPP, data = nas1982)
std <- BetaHC(object)
vcov(std)
```

## vcov.essandwich *Sampling Covariance Matrix for an Object of Class* essandwich

### Description

Sampling Covariance Matrix for an Object of Class essandwich

### Usage

```
## S3 method for class 'essandwich'
vcov(object, ...)
```

## Arguments

| | |
|---|---|
| object | Object of class essandwich. |
| ... | additional arguments. |

## Value

Returns a matrix of the variance-covariance matrix of the estimates.

## Author(s)

Ivan Jacob Agaloos Pesigan

## Examples

```
object <- lm(QUALITY ~ NARTIC + PCTGRT + PCTSUPP, data = nas1982)
std <- BetaHC(object)
out <- RSqSandwich(std)
vcov(out)
```

# Index


∗ **Beta Sandwich Functions**
BetaADF, 2
BetaHC, 4
BetaN, 6
DeltaRSqSandwich, 11
DiffBetaSandwich, 12
FSqSandwich, 13
PCorSandwich, 15
RSqSandwich, 19
SCorSandwich, 20
∗ **betaSandwich**
BetaADF, 2
BetaHC, 4
BetaN, 6
DeltaRSqSandwich, 11
DiffBetaSandwich, 12
FSqSandwich, 13
PCorSandwich, 15
RSqSandwich, 19
SCorSandwich, 20
∗ **data**
nas1982, 14
∗ **deltarsq**
DeltaRSqSandwich, 11
∗ **diff**
DiffBetaSandwich, 12
∗ **fsq**
FSqSandwich, 13
∗ **methods**
coef.betasandwich, 7
coef.essandwich, 8
confint.betasandwich, 9
confint.essandwich, 10
print.betasandwich, 16
print.essandwich, 17
print.summary.essandwich, 18
summary.betasandwich, 21
summary.essandwich, 22
vcov.betasandwich, 23
vcov.essandwich, 23
∗ **pcor**
PCorSandwich, 15
∗ **rsq**
RSqSandwich, 19
∗ **scor**
SCorSandwich, 20
∗ **std**
BetaADF, 2
BetaHC, 4
BetaN, 6

BetaADF, 2
BetaADF(), *3*, *5*, *7*, *11–16*, *19–21*
betaDelta::BetaDelta(), *3*, *6*
BetaHC, 4
BetaHC(), *3*, *6*, *7*, *11–16*, *19–21*
BetaN, 6
BetaN(), *3*, *5*, *6*, *11–16*, *19–21*

coef.betasandwich, 7
coef.essandwich, 8
confint.betasandwich, 9
confint.essandwich, 10

DeltaRSqSandwich, 11
DeltaRSqSandwich(), *3*, *5*, *7*, *13*, *14*, *16*, *19*, *21*
DiffBetaSandwich, 12
DiffBetaSandwich(), *3*, *5*, *7*, *11*, *14*, *16*, *19*, *21*

FSqSandwich, 13
FSqSandwich(), *3*, *5*, *7*, *11*, *13*, *16*, *19*, *21*

nas1982, 14

PCorSandwich, 15
PCorSandwich(), *3*, *5*, *7*, *11*, *13*, *14*, *19*, *21*
print.betasandwich, 16
print.essandwich, 17

print.summary.essandwich, 18

RSqSandwich, 19
RSqSandwich(), *3*, *5*, *7*, *11*, *13, 14*, *16*, *21*

SCorSandwich, 20
SCorSandwich(), *3*, *5*, *7*, *11*, *13, 14*, *16*, *19*
summary.betasandwich, 21
summary.essandwich, 22

vcov.betasandwich, 23
vcov.essandwich, 23